

Use case	Check Train Status	
Release (id)	UC-05, v1.0, 26 April 2026	
Context of Use	A user wants to track a specific train to know if it is on time or to see delay predictions.	
Scope	Allow users to track the real-time location of a specific train and view accurate delay predictions to better organize their journey.	
Level	Primary task	
Primary actor	General user	
StakeHolder & interest	StakeHolder	Interest
	User	Needs accurate and timely updates to organize their journey.
	Prediction System	Must provide reliable data based on current track conditions.
Preconditions	→ The system has an active connection to the Train Management System and Track Sensors.	
Minimal Guarantees	→ If real-time data is temporarily unavailable, the system displays the scheduled timetable data with a warning.	
Success Guarantees	→ The system correctly displays the train's current location, upcoming stops, and an accurate delay prediction.	
Trigger	The user enters a Train ID or route details into the "Train Status" search bar.	
Description (Main Success Scenario)	Step	Action
	1	User inputs the Train ID and requests the status.
	2	System queries the Train Management System and the Prediction System.
	3	System retrieves real-time location and estimated delay data.
	4	System displays the visual timeline of the train, highlighting current position and estimated arrival times for subsequent stops.
Extension	Step	Other action
	2a	<i>The entered Train ID does not exist or the train is not scheduled for today.</i>
	2a ₁	System displays a "Train not found" message.
	2a ₂	System prompts the user to check the ID or search by route.
Technology & data Variations	Relies heavily on the internal Delay Predictor module and live IoT Track Sensors.	